

DIAL — AI, ESTIMATION, PROJECT & TRANSACTION ARCHITECTURE

Second Follow-Up: Detailed Recommended Solutions & Execution Blueprint | August 2026

This document extends the DIAL business-plan and technical-blueprint work with the refined solutions developed around AI job estimation, technician selection, project jobs, protected payments, reconciliation, low-cost AI integration and Hugging Face/open-source AI.

1. Core Design Principle

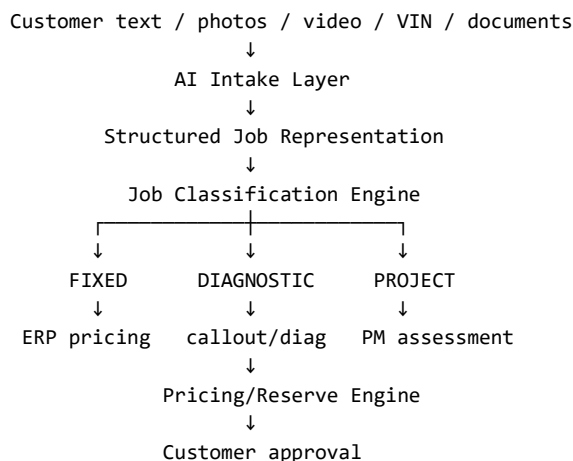
DIAL should operate as a managed marketplace, not a technician directory. The platform manages intake, assessment, matching, protected payment, dispatch, work, reconciliation and quality. AI interprets information and makes recommendations; deterministic ERP rules remain authoritative for prices, payments, commissions, refunds, permissions and settlement.

Use a Job Reserve rather than treating every customer payment as an unconditional deposit. Funds are allocated to the job and reconciled against actual approved costs.

2. Four Job Classes

Class	Workflow	Initial payment	Final scope
Fixed service	Service → price → technician → payment → dispatch	Configured booking/full price	ERP catalogue
Diagnostic / variable	Describe/upload → AI triage → technician → callout → diagnosis → quote	Callout/diagnostic	Technician + DIAL rules
Estimated repair/service	AI assessment → estimate → technician → reserve → confirmation → work	Job reserve	ERP + technician
Project	Request → PM assigned → PM assessment → quotation → approval → DIAL builds team	PM callout only initially	PM + DIAL pricing engine

3. AI Job Estimation Engine



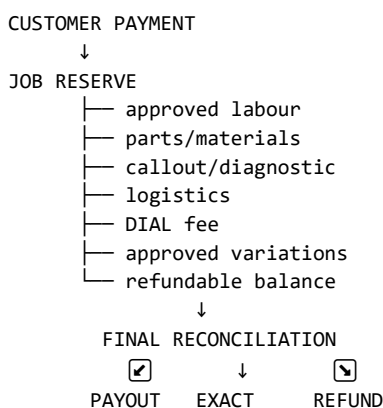
- AI extracts trade, service, symptoms, vehicle, urgency, location and constraints.
- AI asks only questions that materially improve the assessment.
- Photos, video, voice notes, VIN and documents are supported.
- AI returns structured data and confidence, not uncontrolled pricing prose.
- Low-confidence cases are routed to diagnosis/inspection.

4. Confidence-Based Estimation

Confidence	Recommended behaviour
High	Use verified ERP rules/data to produce a fixed or narrow estimate.
Medium	Provide a range and require technician confirmation.
Low	Use diagnostic/callout workflow instead of pretending to know the repair.
Project / complex	Do not remotely promise a final project price; charge PM assessment callout.

AI estimates are forecasts. The DIAL Pricing Engine creates the commercial truth from current catalogue prices, supplier costs, labour rates, callouts, logistics, taxes/fees and DIAL commissions.

5. Job Reserve, Variations & Reconciliation



Example: customer reserves US\$200; final approved cost is US\$150. The ledger records US\$50 as refundable. If diagnosis is performed and the customer declines the repair, configured callout/diagnostic charges and genuinely committed non-refundable costs are retained and the remainder is refundable.

Additional work requires a digital variation showing original scope, additional scope, additional amount and new total. Customer approval is required before non-emergency additional work proceeds.

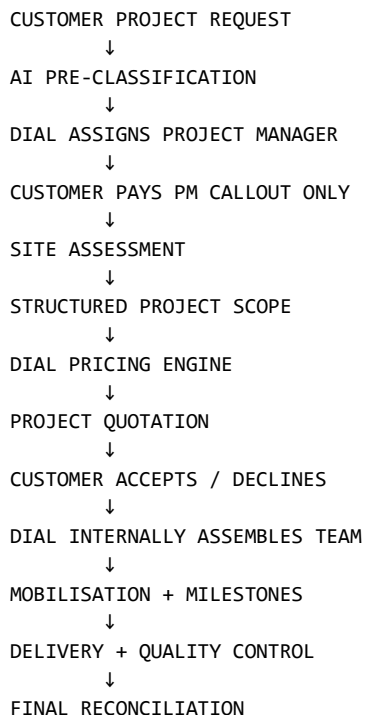
6. When Customers Choose Technicians

Job type	Selection point
Fixed service	After service and price are known; customer chooses from eligible available technicians.
Diagnostic	After DIAL identifies the trade/problem sufficiently; customer chooses from qualified technicians.
Estimated repair	After preliminary scope/skills are known; customer chooses a suitable technician.
Project	Customer does not assemble the team; DIAL assigns a Project Manager and builds the team internally.
Emergency	DIAL can prioritise nearest qualified available technician, subject to customer visibility and configured rules.

Technician cards should show verified trade, relevant experience, completed jobs, completion rate, punctuality, response time, reviews, distance/ETA, availability and professional profile.

7. Project Jobs — No Preset Customer-Facing Teams

Preset teams should not be sold to customers. The customer is buying a project outcome, not selecting an arbitrary bundle of workers. DIAL internally constructs the best team after the Project Manager has assessed the actual work.



7.1 Project Manager assessment

- Customer requirements and acceptance criteria.
- Site measurements and conditions.
- Existing infrastructure and constraints.
- Required trades and quantities.
- Materials/specifications and quantities.
- Labour, worker count and equipment.
- Dependencies, risks and potential variations.
- Timeline and milestones.
- Photographic/site evidence.

7.2 Internal team assembly

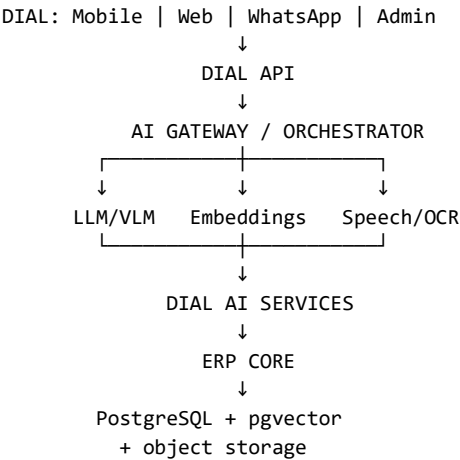
Filter first by trade, verification, qualification, location, availability and equipment. Then rank candidates by relevant experience, reliability, performance, price, workload and similar-job history. The customer can see the assigned professionals for transparency but does not need to construct the team.

7.3 Project payment

After quotation approval, use milestone-based payments rather than one large deposit. Milestones can cover mobilisation, procurement, defined work stages and completion, with evidence and approval rules before release.

8. AI Integration at Minimum Cost

Do not build an AI model from scratch. Build an AI Gateway/Orchestrator that can use commercial APIs, Hugging Face models and eventually self-hosted/fine-tuned models behind one stable DIAL interface.



8.1 Gateway functions

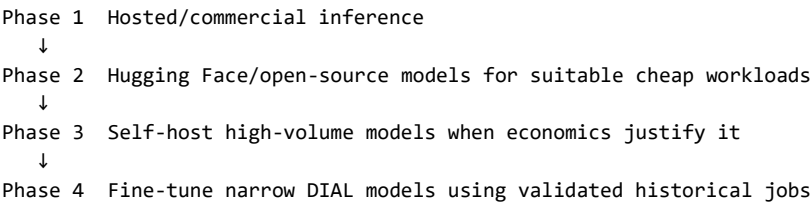
- Provider/model abstraction.
- Structured-output enforcement.
- Prompt/version management.
- Token and cost accounting.
- Rate limiting, retries and fallbacks.
- Caching and deduplication.
- Audit logs.
- AI budget per job/workflow.
- Data/privacy controls.

8.2 Route work by cost

Work	Economical approach
Simple extraction/classification	Small model or deterministic rules
Complex reasoning	Stronger LLM only when needed
Images/video	Vision-language model
Voice	Speech-to-text then normal intake
Documents	OCR/document model + extraction
Semantic search	Embeddings + pgvector
Prices	ERP rules, never free-form AI
Distance/availability	Database/geospatial logic
Technician ranking	Eligibility filters + scoring/ML
Anomaly detection	Rules/statistical ML + human review

9. Hugging Face / Open-Source Strategy

Hugging Face can reduce long-term inference costs and enable customisation. It should be one provider option inside the AI Gateway, not the foundation of the ERP.



Do not fine-tune prematurely. Start with general/open models plus DIAL knowledge retrieval. Once DIAL has enough validated examples, fine-tune focused models such as job classification or structured intake rather than attempting a single general-purpose DIAL model.

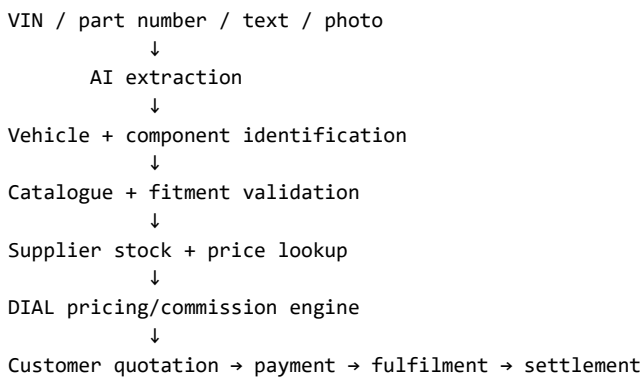
10. RAG, PostgreSQL & pgvector

Use Retrieval-Augmented Generation to give models DIAL-specific knowledge without training it into the model. PostgreSQL plus pgvector is a cost-effective initial vector layer.

- Service procedures and catalogue.
- Vehicle/parts information.
- Historical completed jobs.
- Troubleshooting documents.
- Technician skills and verified experience.
- Project assessment templates.
- Supplier/catalogue descriptions.
- DIAL policies.

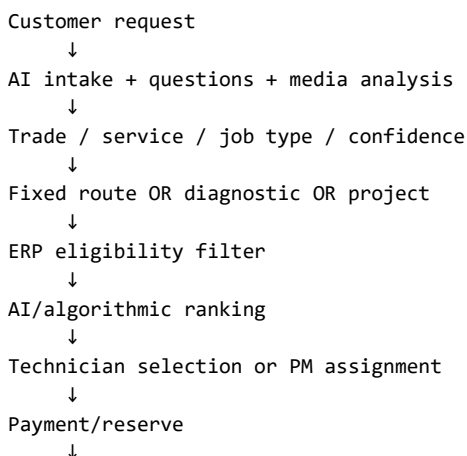
ERP records remain authoritative. RAG supplies context; it does not replace the source-of-truth database.

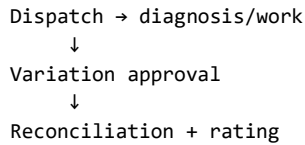
11. Dial a Spare AI Workflow



- Supplier identity can remain hidden from customers.
- Equivalent supplier stock can be represented by product/brand and DIAL-controlled identifiers.
- AI never invents stock, fitment or supplier price; ERP data validates it.

12. Dial a Tech AI Workflow





13. Customer & Technician Protection

- Every booking receives a unique job ID.
- Customer funds remain allocated to the job until earned/refunded.
- Technician arrival/departure and work evidence are timestamped.
- Scope changes require digital approval.
- Customer and technician can submit evidence for disputes.
- Ratings separate workmanship, professionalism, punctuality, communication and completion.
- Branded workwear can be a professional-conduct requirement.
- Disputes use an evidence-based human review process.

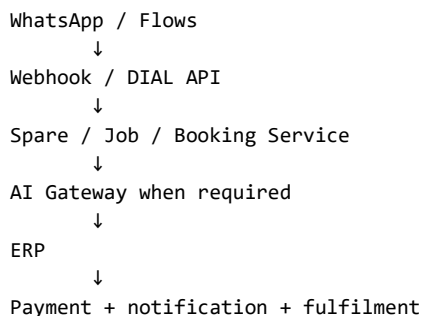
14. Commission Protection & Anti-Collusion

Commission protection should be engineered into the workflow rather than relying only on contracts.

- Every job, technician assignment, customer approval and payment is linked to a unique job record.
- Final payment, completion and rating are tied to the DIAL job.
- Technicians cannot close jobs without required evidence.
- Monitor repeated customer-technician interactions, unusual cancellations, quote changes and off-platform patterns.
- Use contractual consequences and suspension/review for confirmed circumvention.
- Do not automatically accuse users based solely on an AI anomaly score.

15. WhatsApp as an Additional Sales Channel

WhatsApp is an interface into the same DIAL backend, not a separate system.



Dial a Spare: VIN/part number → catalogue lookup → options → order/payment. Dial a Tech: trade → eligible technician cards → selection where applicable → booking fee/reserve → confirmation. Mobile, web and WhatsApp use the same jobs, customers, payments and catalogue.

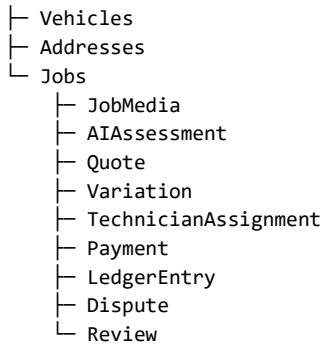
16. Recommended ERP Service Structure

Service	Role
Customer	Profiles, addresses, vehicles
Job	Lifecycle, scope, media, status
AI	Gateway, assessment, structured outputs
Pricing	Rates, rules, quotes, variations
Technician	Profiles, verification, skills, availability

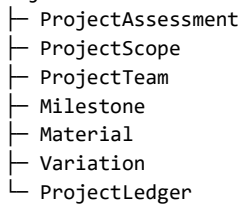
Matching	Eligibility, distance, ranking
Project	PM assessments, teams, milestones
Catalogue	Parts/services, fitment, product data
Supplier	Feeds, stock, costs, settlement
Order	Orders, fulfilment, returns
Payment	Payment intents, capture, refunds
Ledger	Allocations, commissions, payouts
Dispatch	Location, ETA, status
Quality	Ratings, performance
Dispute	Evidence and resolution
Notification	Push, SMS, WhatsApp, email

17. Recommended Data Structures

Customer



Project



AI records should store model/provider, prompt version, input references, structured output, confidence, cost, timestamp and human corrections. This creates an audit trail and future training dataset.

18. Cost Controls

- Use deterministic ERP rules before AI wherever possible.
- Use the smallest adequate model.
- Cache repeated analysis.
- Store conversation state rather than resending full histories.
- Process heavy analysis asynchronously through a queue.
- Set per-job AI budgets.
- Measure cost per workflow.
- Use embeddings for retrieval instead of repeatedly sending long documents.
- Self-host high-volume open-source models when API savings justify infrastructure.
- Benchmark quality versus cost before changing models.

19. Human-in-the-Loop

Decision	Automation
Simple classification	High

Catalogue extraction	High, with validation
Preliminary estimate	AI recommendation
Final repair diagnosis	Technician confirmation
Project scope	Project Manager confirmation
Final price	ERP pricing rules + approved scope
Refund/payout	ERP/payment rules
Dispute	Human review supported by AI evidence

AI should never be the final authority over money or professional liability.

20. Implementation Roadmap

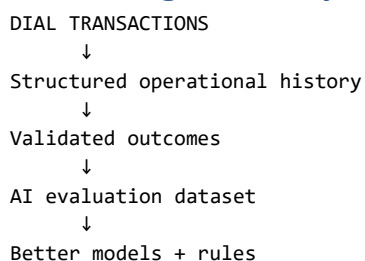
1. Build ERP foundations: identity, customers, technicians, jobs, catalogue, orders, payments and ledger.
2. Build the provider-neutral AI Gateway with one initial model provider.
3. Implement AI job intake and classification.
4. Implement fixed/diagnostic/project routing.
5. Implement deterministic Pricing Engine, Job Reserve and reconciliation.
6. Implement technician availability, eligibility and customer selection.
7. Implement Project Manager callout and structured project assessment.
8. Implement internal project team assembly and milestone payments.
9. Add vision, OCR and speech only where they solve real workflows.
10. Add WhatsApp using the same backend APIs.
11. Add pgvector/RAG and historical-job retrieval.
12. Record AI prediction → human correction → actual outcome.
13. Benchmark Hugging Face/open-source models and self-host appropriate workloads.
14. Fine-tune narrow DIAL models only after enough validated data exists.

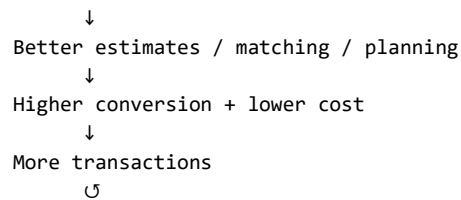
21. MVP Boundary

- AI job intake.
- Fixed-price service workflow.
- Diagnostic callout workflow.
- AI-assisted estimate/range.
- Technician availability and selection.
- Project Manager assessment and project quotation.
- Internal project team assembly.
- Job Reserve and reconciliation.
- Catalogue/supplier integration.
- Mobile/web applications.
- WhatsApp as an additional channel.
- AI cost/audit logging.

Start with a limited set of high-volume automotive/spares and technician workflows, prove economics and collect reliable data before expanding to every trade.

22. Strategic Data Flywheel





The long-term advantage is not merely access to AI models. It is the proprietary Zimbabwean operational dataset generated by DIAL's jobs, prices, parts, technician performance, project outcomes, disputes and customer feedback.

23. Final Architecture Rule

AI interprets and recommends. The ERP validates. Qualified professionals confirm uncertain or consequential decisions. The pricing engine determines commercial amounts. The ledger controls money. Customers approve material scope changes. Every completed job becomes structured data that improves the next decision.